

# Viability Validation Report: Agricultural Compliance Platform

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## Executive Summary

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### Viability Assessment: **CONDITIONALLY VIABLE**

The agricultural compliance platform addresses a real \$500M+ TAM with quantifiable customer pain (35-50 hours/season, \$2,600-4,200 value). Expert financial analysis confirms unit economics are sustainable IF critical assumptions validate. However, the go-to-market strategy (consultant partnerships) and retention model (seasonal re-activation) remain unproven and represent execution risk.

**Recommendation:** Proceed with structured 16-week validation sprint. Treat Weeks 1-4 (consultant partnerships) as existential go/no-go gate. If 2+ consultants commit with favorable terms (<20% revenue share), continue to MVP and pilot phase. If consultant adoption stalls, pivot before burning 12 weeks on product development.

**Funding Viability:** \$300-400K seed is achievable IF you demonstrate traction (5-8 paying customers + 20-30 pilot applicators) by Month 4. Without traction, pursue AgTech grants or revenue-based financing instead.

## Financial Viability Assessment

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### Unit Economics (Revised, Conservative)

Monthly Revenue (20-30 Year 1 customers): \$2,000-3,000 MRR Year 1 Total Revenue: \$150K-200K (ramping) Monthly Burn Rate: \$20-40K (including founder salary, dev team, pilot support) Year 1 Net Cash Burn (after revenue): -\$90K to -\$140K

**Customer Acquisition Cost: \$400-700** - Pilot support: \$200-300/customer (compliance requires hands-on support) - Consultant training/co-marketing: \$100-200/customer amortized - Blended CAC: \$400-700 (revised from \$200-500)

**Lifetime Value (3-year horizon with 50% re-activation): \$2,000-2,500** - Monthly revenue: \$100/month average - Year 1 re-activation: 50% (seasonal dormancy Nov-March) - Year 2+ retention: 70% (if re-activation succeeds)

**LTV:CAC Ratio: 3-5x (Sustainable, barely)**

Viability hinges on achieving 50%+ re-activation and maintaining CAC below \$700. If consultant revenue share exceeds 20%, unit economics break.

## Go/No-Go Conditions (Next 16 Weeks)

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### Required Conditions for Seed Fundraising:

1. **Consultant Partnerships (Weeks 1-4):** 2+ regional consultants commit to pilots with revenue share <20%. If not achieved by Week 4, pivot before MVP build.
2. **MVP Development (Weeks 5-8):** No-code MVP built and pilot-ready by Week 8, prioritizing contractor management feature (differentiation vs. SprayMapper).
3. **Pilot Recruitment (Weeks 9-12):** 20-30 applicators recruited and onboarded via consultant introductions, with founder/team hands-on support.
4. **Conversion Validation (Weeks 13-16):** Preliminary data showing 15%+ pilot-to-paid conversion rate and 50%+ re-activation intent from pilot cohort.

If all four conditions are met by Week 16, you have credible seed pitch. If any condition fails, diagnose root cause and pivot strategy before fundraising.

## Critical Assumptions & Risks

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### Yellow Flags (Validate Immediately):

1. **Consultant Active Promotion (HIGH RISK):** You haven't confirmed consultants will actively recommend the tool. Passive acceptance ('I'll mention it if asked') is insufficient for customer acquisition targets. Script question: 'Would you recommend this to every applicable client?'
2. **Re-activation Rate at 50%+ (HIGH RISK):** Seasonal dormancy (Nov-March) is different from churn, but re-activation requires engagement strategy. Without push notifications, email campaigns, or rule updates during off-season, applicators may forget the tool.
3. **CAC Realism at \$400-700 (MEDIUM RISK):** Pilot support costs (\$200-300) are higher than Canvas assumption (\$100). If conversion rate is 15% instead of 25%, effective CAC rises to \$900+, breaking unit economics.
4. **Revenue Share Expectations (HIGH RISK):** If consultants demand 20-30% commission on referred customers, CAC increases to \$1,000+ per acquisition, making model unsustainable.

### Red Flags (Monitor, Plan Contingency):

1. **Competitive Response (Month 6-9):** SprayMapper mobile launch could neutralize offline

differentiation. Mitigation: Prioritize contractor management feature as unique defensibility.

2. **Year 1 Customer Target (Realistic: 5-8, not 20-30):** Consultant channel delivers slower than optimistic projections. Set seed investor expectations accordingly.
3. **Seasonal Churn/Re-engagement:** Off-season dormancy requires active re-engagement (emails, feature updates, rule changes) to prevent churn during re-activation phase.

## 16-Week Validation Roadmap

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**Weeks 1-4: Consultant Validation Sprint (GO/NO-GO GATE)** - Contact 10 SE Missouri ag consultants; schedule discovery calls - Ask: 'Do your clients struggle with compliance? Would you recommend a tool?' - Negotiate: Revenue share model, referral fees, pilot logistics - Goal: Lock in 2 consultants committing to 5-10 applicator introductions each - Decision: If 2+ committed by Week 4 with <20% rev share, proceed. Otherwise, pivot.

**Weeks 5-8: MVP Build (7-Week Compression)** - Use no-code (Figma mockups + Airtable backend or similar) - Prioritize: Mobile interface, EPA RUP database integration, contractor management feature - Ship: Pilot-ready version by Week 8 (not polished, but functional)

**Weeks 9-12: Pilot Launch (20-30 Applicators)** - Onboard via consultant introductions; founder/team provides white-glove support - Measure: Daily active users, feature usage, compliance accuracy feedback - Track: Willingness to pay (ask at Week 10), likelihood to renew (Week 11)

**Weeks 13-16: Analysis & Decision** - Calculate pilot conversion rate (target: 15%+ say they'd pay) - Measure re-activation intent (target: 50%+ say they'd use next season) - Recalculate CAC, LTV, unit economics with real data - Decision: Seed pitch if metrics validate. Otherwise, pivot strategy.

## Seed Pitch Narrative (Month 4)

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### Frame for Investors:

'We identified a fragmented \$500M+ compliance software market where applicators spend 35-50 hours/season on manual record-keeping, creating regulatory risk and operational inefficiency. We built a mobile-first MVP with offline capability and contractor management (unmet need identified in customer research). We tested with 20-30 applicators via trusted ag consultant partnerships. Preliminary traction: 15-20% pilot conversion, 50%+ re-activation intent. CAC is \$400-700; LTV:CAC is 3-5x. We're raising \$300-400K for 18-month runway to scale from 5-8 Year 1 customers to 50-80 Year 2 customers via consultant network expansion and word-of-mouth. Competitive differentiation: contractor management and first-mover regional distribution advantage before SprayMapper responds (expected Month 6-9).'

**Key Metrics to Share:** - Actual pilot conversion rate from real applicators - Re-activation intent data (% saying they'd re-engage next season) - Consultant partnership commitments (2+ named, with referral commitments) - CAC measured from actual acquisition costs - Retention assumptions grounded in seasonal model

## Recommended Actions (Immediate)

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**This Week:** 1. Create list of 10-15 SE Missouri ag consultants (State Extension, crop advisors, independent consultants) 2. Identify warm intros; schedule first 5 discovery calls 3. Prepare consultant pitch deck (problem, solution, referral model)

**Week 1-2:** 1. Complete 5-10 consultant discovery calls 2. Filter to top 3-5 based on applicator client base and compliance pain point frequency 3. Schedule deep-dive negotiations with top 3

**Week 2-3:** 1. Present MVP concept to top consultants (Figma mockup, not code) 2. Explicitly ask: 'Would you recommend this? What's your referral model?' 3. Get 2 commitments or identify why adoption is stalling

**Week 4:** 1. Lock in consultant agreements with written commitments 2. Plan pilot logistics (applicator introductions, timeline, support model) 3. **GO/NO-GO DECISION:** Proceed to MVP build or pivot strategy

**Ongoing (Weeks 5-16):** - Weekly measurement of pilot metrics (conversion, engagement, re-activation intent) - Monthly founder check-ins with consultant partners - Explicit tracking of CAC, LTV, and retention assumptions vs. reality